

## Peer Review Comments and Discussion (Daniel Egger, Duke U.)

Comparing total profitability over the test set (but applying each model as fit to the training set) is the best approach available for determining whether to approve a credit card application or not.

### Binary Classification Comments

Total profitability on the test set is calculated for the binary classification model simply by adding the profits and losses for all individuals who would have been issued credit cards under the model.

In other words, add the actual, given profits and losses for all “Test Negatives” – True Negatives and False Negatives.

All the binary models are at a big disadvantage over the new linear model when trying to optimize profits – because profits and losses turn out not to be highly correlated with default.

[Note: Even the Eggertopia model, which appears better than most models constructed from the six input variables when evaluated by AUC or the “old” cost estimates of the bank, performs quite poorly if the original threshold of 0.1 is used on the test set – capturing bank profits of only \$51,382.)

### New Cost per FP and Cost per FN values

An important point to consider is that these new profitability data allow us to replace the bank’s “old” estimate of \$2,500 profit per non-defaulting customer and \$5,000 loss per defaulting customer.

Therefore, while keeping your original model, you *could* change the threshold on the training set to better optimize costs.

New values are:

|                                                          | Training Set    | Test Set        | Mean            |
|----------------------------------------------------------|-----------------|-----------------|-----------------|
| <b>Profits per non-default<br/>(Same as cost per FP)</b> | <b>+\$3,799</b> | <b>+\$3,248</b> | <b>+\$3,523</b> |
| <b>Losses per default<br/>(Same as cost per FN)</b>      | <b>-\$3775</b>  | <b>-\$3612</b>  | <b>-\$3694</b>  |

Once you’ve applied either your own threshold or the new, adjusted threshold from the training set to the test set, add the profitability values for all “test negatives” on the test set to determine the total profitability of the binary model.

*However*, most binary models – because they were *originally* designed to capture default/non-default signals rather than the new profitability signals – will still not perform as well as the linear regression model.

## Linear Regression Model

For the linear regression model to yield a total profit number and be compared to the binary model, a new, additional decision will still need to be made whether to issue each applicant a card or not. So some kind of binary classification will need to be added to the linear model.

The linear regression model gives a point estimate of profitability and a range for 90% confidence in the estimate. The 90% confidence range is over \$12,000 (\$6,168 above and below each estimate) and the average profit per customer is only \$1,600. This suggests the regression model is not very precise for individual forecasts. But by the Central Limit Theorem, the sample mean of residuals for a large group of forecasts will contain some new error but nevertheless be far more accurate than individual forecasts.

Therefore, a good approach would be to choose all applicants whose *point estimate* of profitability on the *test set* is non-negative.

Using the \$5,755 standard deviation and \$1,905 average from the training set would be the correct way to estimate profitability on the test set.

Setting a binary threshold between profitable-or-breakeven and unprofitable applicant forecasts would lead to including 115 applicants – and a total bank profit on the test set of \$365,908.

Another approach is to exclude anyone who has a certain percent chance of losing money for the bank. For example, it would be acceptable to exclude anyone who had a 25% probability or higher of generating losses. However, because the error on the linear regression is quite large, this approach will exclude more people than the linear regression model. While the profit per customer may be higher, the total profits will be lower.

A third approach would be to pick the 75% (or other percentile) of applicants with the highest projected profitability. Choosing the top 75% by estimated profitability would lead to a bank profitability on the test set of \$407,233.

This compares very well to the “hindsight” profit on the test set – the profit achievable with complete information about future profitability - \$554,797.

Hybrid approaches can be considered – although not explicitly addressed in the question. For example, the binary model can be used to select non-defaulters, and then classified defaulters can be reviewed again to see if they fall in a potential high profitability category under the linear regression model, in which case they can be “added back” to the group issued credit cards.

Note that if the identical linear regression “betas” on standardized data are used to construct a single score for the binary classification, the resulting rank-ordering will *not* be optimized to maximize AUC for default. However, it will be optimized to provide the *same* result of \$365,908 above.